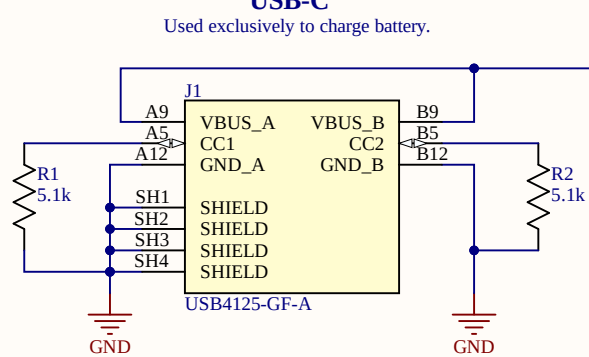


USB-C

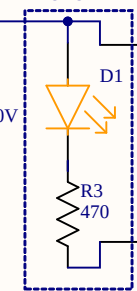
Used exclusively to charge battery.



USB-C plug will be placed on edge of the PCB for access through housing.

Power Charger

Charging Status

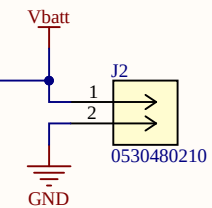


For best thermal performance, add vias from land area of EP to copper layer on opposite side of PCB

Power charger should be placed close to battery connector.

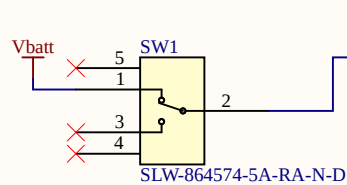
Li Ion Battery

LP384260JU+PCM+MOLEX 51021-0200 35MM



Connector will be placed on edge of the PCB to connect to battery located underneath.

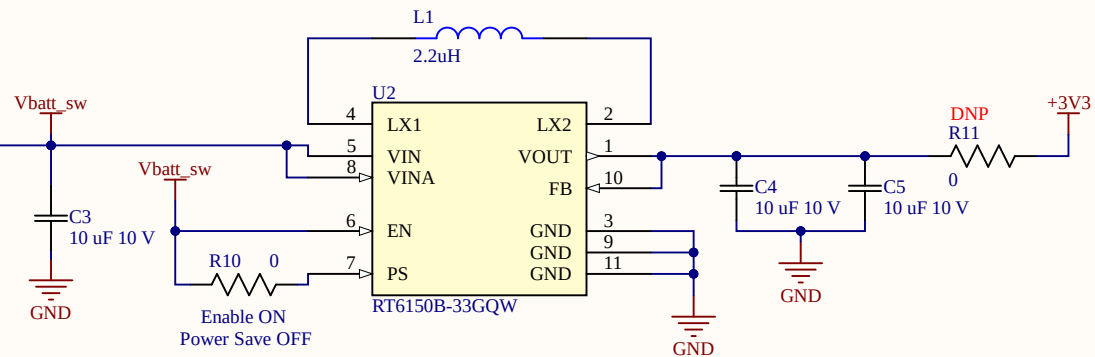
Power Switch



Switch in L position: ON
Switch in R position: OFF

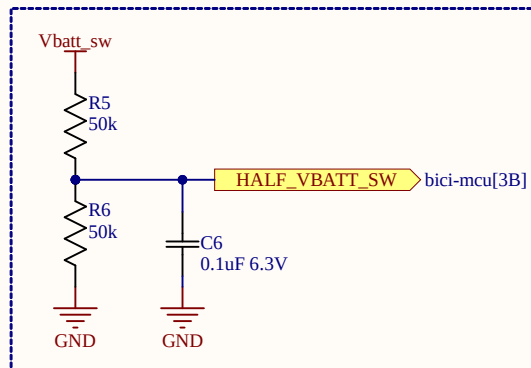
Switch will be placed on edge of the PCB for access through housing.

Buck-Boost

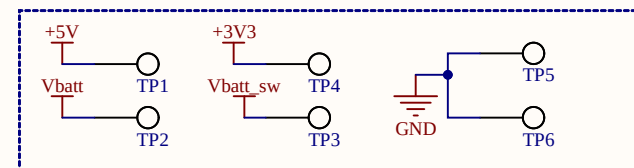


C3 should be placed as close as possible to Vin.
C4 and C5 should be placed as close as possible to Vout.
L1 should be connected to inductor by wide and short trace.

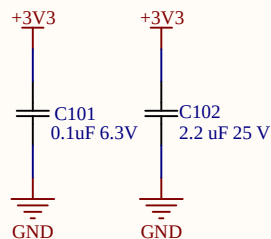
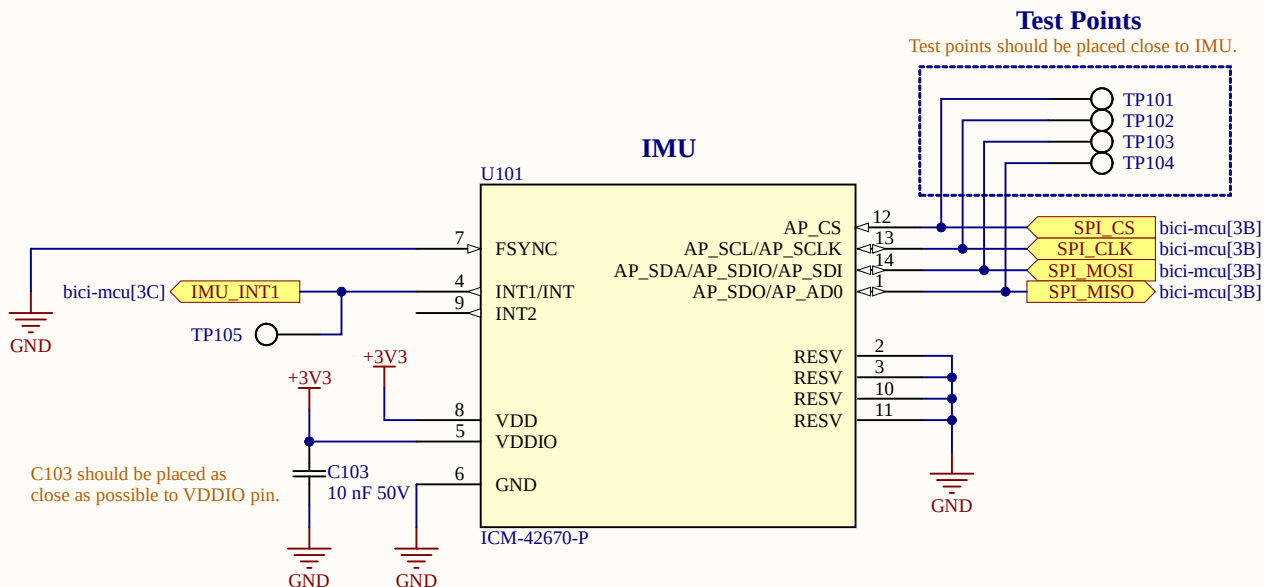
Low Power Detection



Test Points



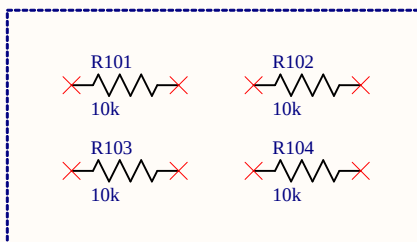
Title Bici: Power Management Circuitry related to system power and charging.		
Size A	Number	Revision V5
Date: 2/25/2026	Sheet1 of 5	
File: C:\Users\...\bici-power.SchDoc	Drawn By: Team Bici	



C101 and C102 should be placed as close as possible to VDD pin, with C101 closest to pin.

Spare 0603 Footprints

(Not on BOM) To make for easier soldering if we need to add a voltage divider (for eg) later.
(Per Mike Kofron's suggestion at Fall final presentation.)



Title Bici: IMU Circuitry related to IMU.		
Size A	Number	Revision V5
Date:	2/25/2026	Sheet2 of 5
File:	C:\Users\...\bici-imu.SchDoc	Drawn By: Team Bici

A

A

B

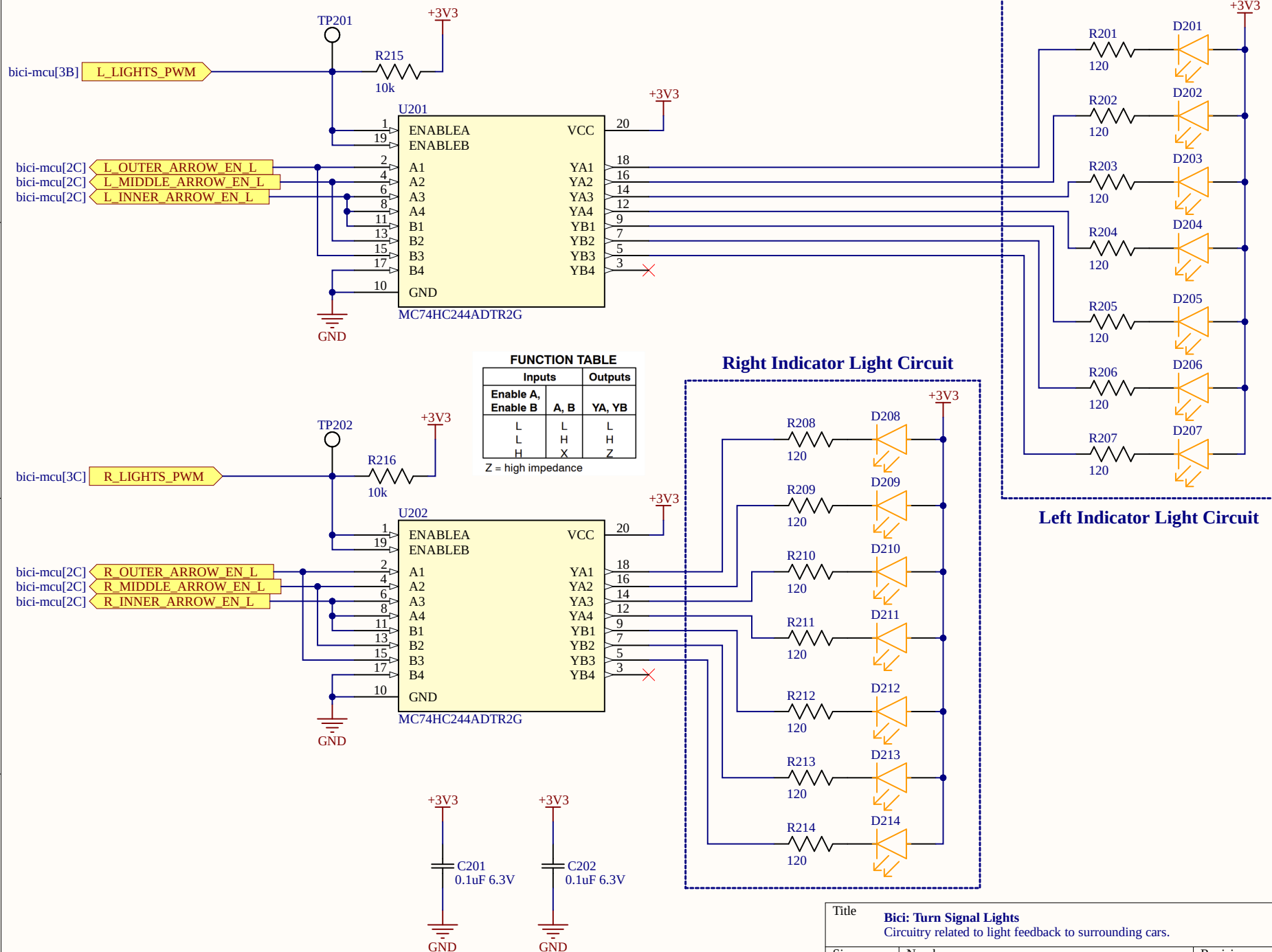
B

C

C

D

D



C202 should be placed as close as possible to U202 VCC pin.
 C201 should be placed as close as possible to U201 VCC pin.

Title Bici: Turn Signal Lights Circuitry related to light feedback to surrounding cars.		
Size A	Number	Revision V5
Date:	2/25/2026	Sheet3 of 5
File:	C:\Users\...\bici-turn-lights.SchDoc	Drawn By: Team Bici

A

B

C

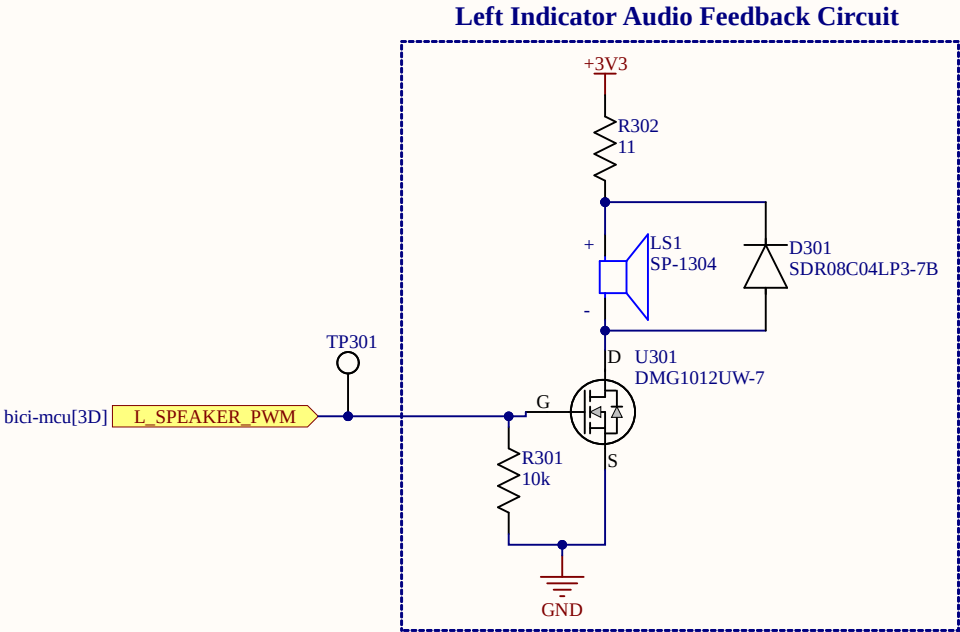
D

A

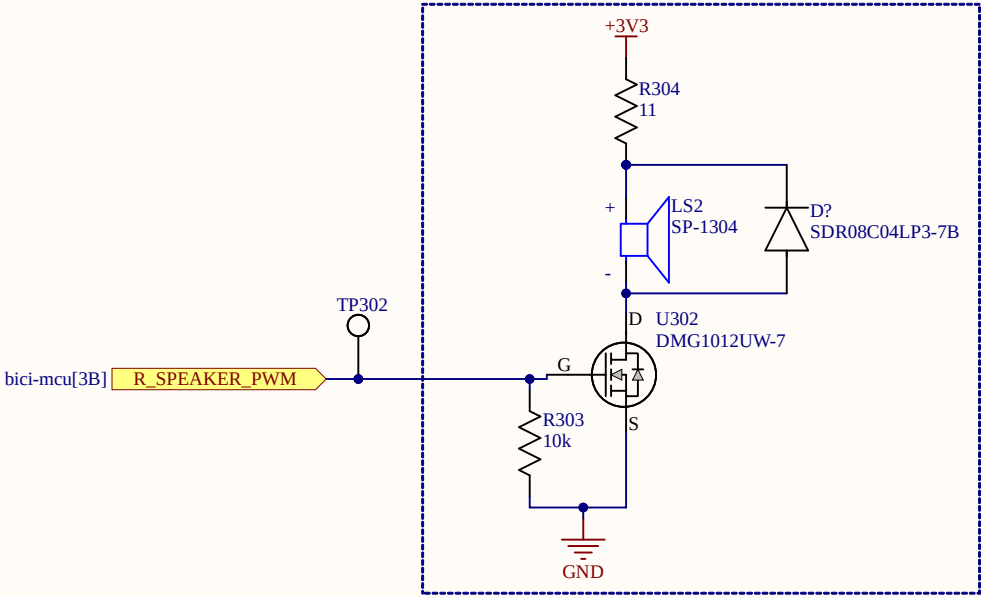
B

C

D



LS1 should be placed at the edge of the PCB on the left arrow point (to accomodate speaker placement in housing).



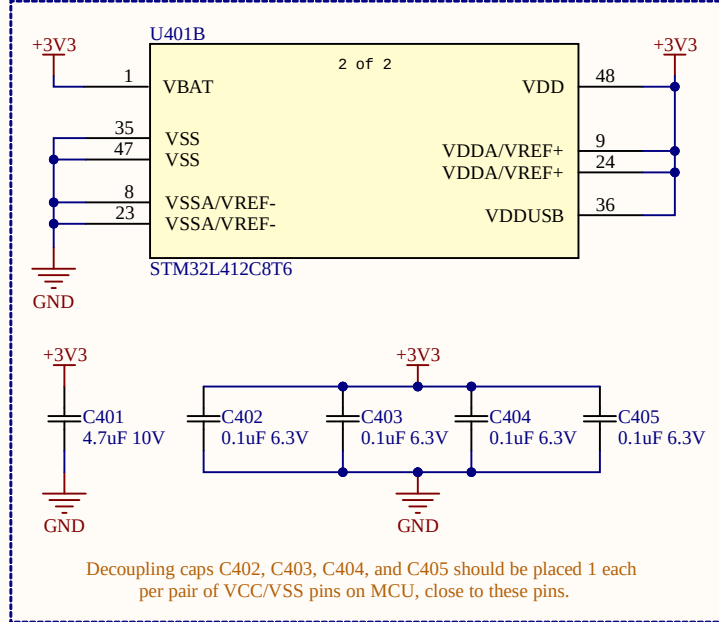
LS2 should be placed at the edge of the PCB on the right arrow point (to accomodate speaker placement in housing).

Right Indicator Audio Feedback Circuit

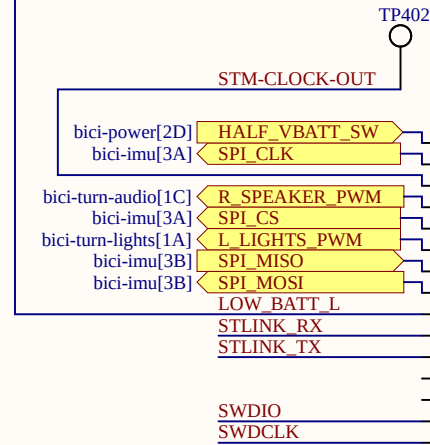
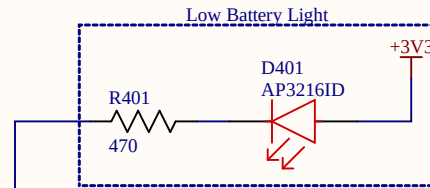
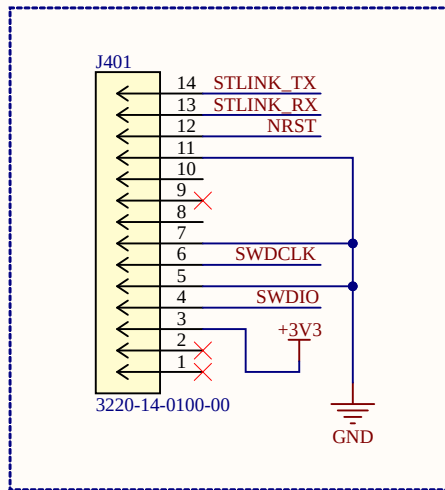
Title			Bici: Turn Signal Audio Circuitry related to audio feedback to user.
Size	Number		Revision
A			V5
Date:	2/25/2026	Sheet4 of	5
File:	C:\Users\...\bici-turn-audio.SchDoc	Drawn By:	Team Bici

STM Power

STM I/O

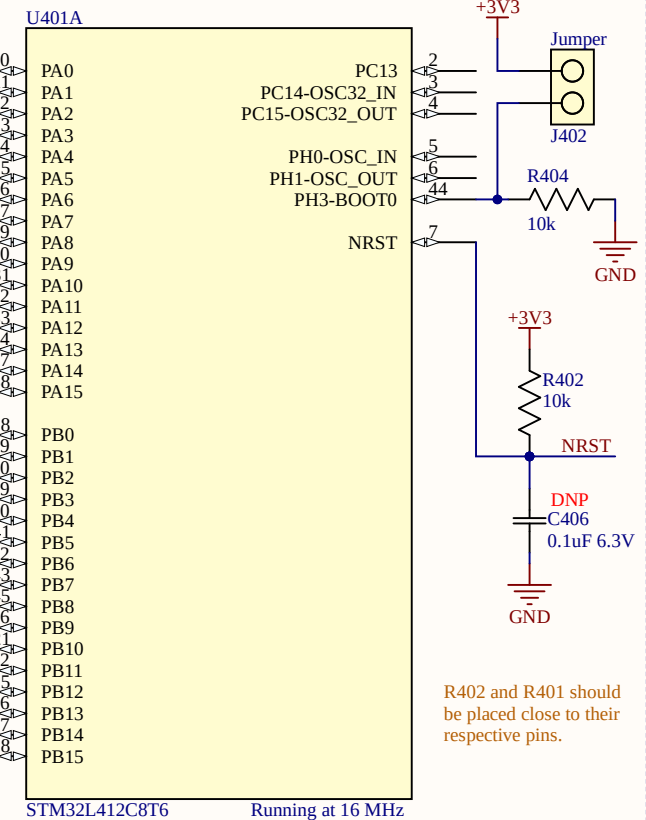


Programming/Debugging Connector



U401A:

- PA0 is ADC in (for low battery detection)
- PA8 is active-low output to actuate low battery light
- PB14 and PA3 are associated with TIM15 for PWM (Audio)
- PA5 and PB3 are associated with TIM2 for PWM (Lights)
- PB0 configured as EXTI0 (external interrupt input) for IMU_INT1
- PA9 is STM USART1 TX
- PA10 is STM USART1 RX
- PA2 is RCC_LSCO for STM clock output
- PB4, PB5, PB6, PB7, PB8, PB9 are active-low enable signals to individually enable the 3 sections of each arrow light
- PB11 and PB12 are spare GPIOs



Title

Bici: MCU
Circuitry related to embedded MCU.

Size

A

Number

Revision

V5

Date: 2/25/2026

Sheet5 of 5

File: C:\Users\...\bici-mcu.SchDoc

Drawn By: Team Bici

